

Computing Machinery and Understanding From the Imitation Game to the Turing: An Operational Measure of Generalized Explanatory Compression

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October 2, 2026

v1.14 preprint (not peer reviewed)

Companion: “Computing Machinery and Understanding: The Turing as a Measure Emerging from AIEN” (v1.7) gives the engineering history the measure emerged from: the AIEN/Omega/FORGE architecture, the public evidence receipts, and the first experimental proposals. This preprint is the formal reference: definitions, proofs, protocol, related work, measurement-validity criteria, and scientific falsification criteria.

Abstract

In 1950, Alan Turing confronted the question “Can machines think?” not by attempting to define thought directly, but by replacing the question with an operational experiment: the imitation game. That methodological move transformed an ill-posed philosophical dispute into an empirical research program.

This paper proposes an analogous reformulation for a contemporary question: **does a machine understand?** I propose that one measurable component of understanding can be operationalized as the discovery of compact structure that continues to predict previously unseen observations. The proposal combines Shannon information, predictive coding, Minimum Description Length, empirical verification, and physical resource accounting.

I define a new derived quantity, **Turing gain**, denoted $\mathcal{T}$. One Turing (1 T)¹ is one bit of **net held-out explanatory compression**: the net reduction in held-out total description length produced by a specified model, program, theory, or abstraction relative to a declared baseline, after including the description cost of the explanatory structure itself. For baseline B , candidate explanation M , sealed held-out observations D , declared shared background G , and measurement profile P ,

$$\mathcal{T}(M; B, D, G, P) = [L(B | G) + L(D | B, G)] - [L(M | G) + L(D | M, G)].$$

Positive gain establishes a lower total description length than the declared baseline on the specified held-out data, under the declared code and background. A negative gain indicates that the candidate’s total description length exceeds the baseline’s: the explanation costs more complexity than the structure it captures, predicts worse than the baseline, or both. A smaller model can lose through worse predictions alone, and positive gain can come from a cheaper representation with identical predictions. Turing gain is therefore permitted to fail: a theory can score negative Turings.

The Turing is not proposed as a replacement for the bit, the joule, the watt, or any SI quantity. It is a protocol-bound measure of **generalized explanatory gain**. The claim is

¹The symbol T denotes the Turing unit proposed here. It is distinct from the SI tesla symbol; context and the $\mathcal{T}_{\text{ideal}}$ and $\mathcal{T}_{\text{real}}$ subscripts disambiguate.

deliberately narrow: the Turing measures one operational component of understanding (its predictive and explanatory-efficiency dimension), not understanding itself. It does not measure causal direction, semantic grounding, agency, or value alignment; Section 9 states these exclusions explicitly. When paired with physical measurements, it permits derived quantities such as the Turing rate ($\mathcal{T}/s$), the specific Turing gain ($\mathcal{T}$ per held-out observation), and Turing yield ($Y_{\text{eval}}, Y_{\text{discovery}}$), allowing the informational productivity of different algorithms, architectures, processors, accelerators, and potentially non-digital substrates to be compared under controlled experimental conditions. The yield quantities address a current environmental blind spot: the field has no standardized measure of net held-out explanatory compression per joule. The unit’s discipline (declared baselines, sealed held-out data, recomputable evidence, declared shared background) is also an honesty discipline, and it gives alignment research a measurable target: verified understanding rather than performed competence. The limits are stated explicitly: the Turing does not directly measure safety, consciousness, or moral worth.

The proposal is explicitly testable at two levels, and preregistered criteria are given for both: measurement-validity criteria, which decide whether a reported number is a sound measurement under its profile, and scientific falsification criteria, which state what would fail the claim that the statistic is a useful operational indicator of understanding. At the measurement level: if predictive models do not produce corresponding compression gains, if candidate complexity erases those gains, if held-out improvements fail to reproduce, or if physical attribution cannot satisfy conservation constraints, the claimed measurements fail.

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1 Introduction

Turing taught us to replace an unanswerable question with an operational one. I ask whether machines can discover explanations that make unseen reality less surprising.

I propose to consider the question: **can a machine understand?**

The customary approach would begin by defining the words *machine* and *understand*. This is unlikely to be profitable. Turing did not settle the meaning of “thinking” by definition. He proposed an observable game whose outcome could be discussed without agreement on the metaphysics of thought [1]. I attempt a similar substitution. Instead of asking “does this machine understand the world?”, I ask:

Has this machine discovered a compact explanation that makes previously unseen parts of the world less surprising?

This question admits measurement. A predictive model implicitly defines a code: for a sequence $D = (x_1, \dots, x_n)$,

$$L(D \mid M, G) = - \sum_{t=1}^n \log_2 p_M(x_t \mid x_{<t}, G)$$

is its idealized predictive description length under model M given the declared shared background G (Section 5.7), where $p_M(x_t \mid x_{<t}, G)$ is the probability the model assigns to observation x_t given the preceding observations and the background. Shannon’s theory established the mathematical relationship between probability, information, and the limits of lossless coding [2]: a model that predicts a sequence more accurately can, in principle, describe that sequence using fewer bits. Thus prediction $\longleftrightarrow$ probability $\longleftrightarrow$ description length. The surprising event is expensive; the predictable event is cheap; regularity is compressible.

But compression of data alone is insufficient, because a machine could memorize an enormous lookup table: excellent predictions, uneconomical explanation. The model itself must also be paid

for. The relevant quantity is the two-part total description length $L(M \mid G) + L(D \mid M, G)$, in the Minimum Description Length tradition [5, 6]. A lookup table costing eight million bits that reduces residual prediction length by only three million bits has a negative net contribution; a 400-bit rule that reduces unexplained held-out data by 50,000 bits has discovered something compact and reusable.

1.1 What the Turing is not: a renamed bit

An immediate objection deserves an answer before the formalism: is the Turing “simply renaming bits”? It is not, and the distinction is load-bearing for everything that follows. Three things must be kept distinct:

- **Quantity.** Turing gain, $\mathcal{T}(M; B, D, G, P)$: the net held-out explanatory advantage of candidate M over baseline B on sealed observations D , conditional on declared shared background G , under measurement profile P (Definition 2).
- **Reporting unit.** The Turing, T : one bit of net held-out explanatory advantage under a declared protocol.
- **Measurement.** A reported value together with its profile: e.g., 84,220 T under profile P . A bare number without its profile is scientifically incomplete.
- **Rate reporting.** The headline number is the **specific Turing gain**: $\mathcal{T}$ divided by the number of sealed held-out observations, in Turings per observation. Raw totals are compared only at fixed sample size. A bare total without its n is as incomplete as a bare number without its profile.

Dimensionally, Turing gain is measured on a bit scale. Scientifically, it is a constrained statistic, not arbitrary information content: the difference between two controlled total description lengths, with the candidate’s own complexity charged, the data sealed, the baseline declared in advance, and the physical attribution conserved. Giving a particular operational meaning to a derived quantity is standard scientific practice. Haldane’s darwin [8] did exactly this for evolutionary rate. It is not the discovery of a new fundamental SI dimension. The Turing claims no such dimension. It claims a disciplined way of counting.

1.2 Turing’s own precedent: the ban and the deciban

The name is not decoration. At Bletchley Park, Turing needed a working unit for the weight of evidence, and he coined one: the *ban*, the base-10 log-likelihood-ratio unit, with the *deciban* as its tenth, named by explicit analogy with the decibel [16]. Cryptanalysts scored hypotheses in decibans because evidence had to be added, compared, and acted on, and that required a unit. The Turing proposed here is the direct descendant of that move: one bit of net held-out explanatory advantage, scored under a declared protocol, so that claims about machine understanding can be added, compared, and acted on. The ban measured evidence for a hypothesis; the Turing measures evidence that an explanation has earned its keep.

1.3 Benchmarks measure performance; the Turing measures explanatory gain

Contemporary machine learning evaluates itself chiefly through benchmarks: fixed test sets on which systems compete for accuracy. This culture has produced real progress and real pathologies. Web-scale pretraining corpora swallow test sets, so “held-out” is often a fiction: contamination must

now be measured per benchmark [39], models reconstruct missing benchmark items at rates far above chance [41], and training data is extractable verbatim [37]. Bender and Koller [34] dramatized the deeper issue with an octopus that predicts telegraphic text superbly while possessing no model of the world the text describes: excellent string prediction with zero understanding. Bender et al. [36] named the resulting systems stochastic parrots: fluent stitchers of linguistic form without reference to meaning. McCoy et al. [43] showed empirically that large language models exhibit probability-shaped rather than understanding-shaped error patterns: GPT-4 decoded a simple cipher at 51% accuracy when the answer was a high-probability sentence versus 13% when it was low-probability, on a deterministic task where probability should not matter.

The fair counter must be stated. Li et al. [38] showed that a transformer trained only on Othello move sequences developed a nonlinear internal representation of board state, and that interventions on that representation causally changed its legal-move predictions: prediction pressure *can* induce causally efficacious world models. Vafa et al. [44] replied that decodable probes do not establish globally coherent, causally usable models. Both can be true: prediction can induce models, but predictive success does not guarantee them.

The Turing sits on this fault line by design. It rewards predictive improvement after accounting for model-description cost, which is the compression leg of understanding. The causal-usability leg must be measured separately, through interventional held-out regimes (Sections 5.4 and 5.6), and the paper claims nothing beyond the leg it measures. What the Turing refuses is the benchmark culture’s central conflation: a score on a possibly memorized test is not evidence of understanding. A positive Turing gain is evidence of exactly one thing: the candidate found structure that pays for itself on data it never saw.

1.4 The arc: from prediction pressure to physical law

Prediction-error minimization is not an arbitrary choice of objective. It is the design principle nature itself uses. Friston [25] argued that organisms minimize variational free energy, an upper bound on sensory surprisal, by updating generative models and by acting to change what they sense; Clark [28] developed the predictive-brain synthesis. Kirby, Cornish, and Smith [23] showed in the laboratory that when languages pass through a learnability bottleneck across generations of learners, they become more structured: “design without a designer,” with compressibility pressure generating the very structure we associate with understanding. Delétang et al. [40] demonstrated the engineering counterpart: large language models are powerful general-purpose compressors, with Chinchilla 70B beating PNG on images and FLAC on audio, because prediction and compression are two faces of one quantity.

But prediction pressure alone is not enough; it must be aimed at the right world. A system trained only on human text inherits humanity’s signal with all of its noise: form without meaning, in Bender and Koller’s phrase. The position of this paper is that the epistemic foundation of machine understanding should be physical law, not human text. Physics-informed neural networks [31] are the flagship demonstration: embedding partial-differential-equation residuals directly into the loss constrains the model by nature’s regularities, so its held-out compression must track those regularities rather than textual correlations. Sutton’s Bitter Lesson [32] supplies both the alliance and the warning. The alliance: general methods that leverage computation beat hand-built human knowledge in the long run, which is an argument for letting a neutral arbiter, held-out compression, decide. The warning: do not privilege human-legible explanations, since the profile should let simple scalable predictors compete on equal footing with elegant theories and let the bits decide.

The thesis, stated plainly: **build what the machine knows on physics, the foundation of all existence; keep purpose, values, and moral direction a human responsibility.**

The Turing measures the first half: the epistemic dimension, how much of nature’s regularity the machine has captured per unit of explanation cost. It is silent on the second half by design (Section 9). EXP-002, the Stochastic Law Discovery experiment, is the concrete test of the first half: the machine must discover stochastic physical structure from sealed trajectories, not imitate text about that structure.

2 Why This Measure Matters

The formalism below is motivated by three concerns. They are stated here as motivation and interpretation, not as substitutes for rigor: the definitions, lemmas, protocol, measurement-validity criteria, and scientific falsification criteria that follow stand or fall on their own. And the limits of Section 9 apply throughout. In particular, the Turing does not directly measure safety, consciousness, or moral worth.

2.1 Environmental safety: a blind spot with a measure

Computation does not occur outside physics, and machine intelligence is scaling its energy consumption faster than the ability to account for what that energy buys. The field has mature metrics for performance per watt, but no standardized measure of net held-out explanatory compression per joule. That is an environmental blind spot: without a measure of informational productivity, there is no principled way to prefer the algorithm that discovers the same structure for a tenth of the energy, and no way to make that preference legible to anyone outside the laboratory.

Turing yield (Definition 4) is proposed as that measure. It does not measure carbon emissions, ecological harm, or the full lifecycle cost of hardware; it measures verified net held-out explanatory compression per measured joule, which is the quantity a researcher or operator can actually control when choosing algorithms, architectures, and substrates. Hypothesis H5 of the experimental protocol (Section 5) is the environmental hypothesis in falsifiable form: semantically equivalent realizations should sometimes exhibit materially different Turing yields. If that hypothesis holds, Turing yield becomes an instrument for steering the field toward informational productivity per joule: the quantity that must improve if machine understanding is to scale within physical limits.

2.2 Humanity: an honesty discipline

The unit’s rules are an honesty discipline. Baselines are declared before measurement, not chosen after. Held-out data is sealed before the candidate is constructed or selected. Shared background is declared, not smuggled: every pretrained model, corpus, and prior artifact in the causal chain of producing the candidate is versioned in the profile (Section 5.7), so a candidate cannot rent a gigantic undeclared model and bill the protocol for the savings. Every number must be recomputable by a stranger from published evidence. And theories are permitted negative Turings: a candidate that costs more complexity than the structure it captures scores below zero, in public, by construction.

This rewards verified explanatory gain over fluent performance. A system optimized for Turings cannot coast on pure memorization: under Lemma 2’s assumptions, a pure memorizer, explicitly charged, cannot earn positive Turings against a cheaper equal default on held-out data. That discipline is narrow by design. It does not forbid stored structure that earns its cost on held-out observations; a table-based predictor can legitimately generalize, and the first reported candidate (Section 7) is reported as one, within the limits stated there (internally replayed; independent blinding not established). A system also cannot cherry-pick its baseline or hide its failures, because

the measurement procedure exposes them as negative scores. A culture that can measure verified explanatory gain can demand it; a culture with only benchmark leaderboards gets what it measures. The Turing is a small piece of infrastructure for the former.

2.3 Alignment: a measurable target

Alignment research asks, among other things, whether a system’s apparent competence reflects genuine understanding or fluent performance. That question is hard to aim at because “understanding” has had no operational measure. The Turing operationalizes one measurable component of understanding: the discovery of compact structure that continues to predict unseen observations, and thereby gives alignment work something quantitative to target and to audit.

The mechanism is the complexity charge: a system that must pay for its own complexity, in the open, on sealed data, faces a measured price for faked understanding. Memorized performance is exposed by $L(M \mid G)$ (Lemma 2); uncalibrated confidence is exposed by $L(D \mid M, G)$; double-counted credit is blocked by informational conservation (Definition 6). None of this measures safety or intent directly. A well-calibrated predictor can still be misused, and Section 9 states the limits plainly. What it contributes is a falsifiable evidentiary criterion for one operational component of understanding: a system’s claimed understanding should survive contact with unseen reality after paying for itself.

3 Formal Definitions

Every symbol is defined before use. Numbered definitions are binding: a reported Turing measurement is valid only if each definition’s requirements are satisfied and recorded in the measurement profile.

Definition 1 (Total description length). *Let M be a candidate explanation: a model, program, theory, abstraction, or explanatory structure. Let $D = (x_1, \dots, x_n)$ be a finite sequence of observations over a finite alphabet. Let p_M be a computable predictive distribution exposed by M , assigning $p_M(x_t \mid x_{<t}, G)$ to each observation given its predecessors and the declared shared background G . Then:*

- *G is the declared shared background: the versioned, cost-accounted set of pretrained models, corpora, interpreters, and toolchains available to the decoder (Section 5.7); every codelength is conditional on G .*
- *$L(M \mid G)$ is the length in bits of M under a declared, computable encoding convention (Assumption A1), with the declared shared background G available to the decoder: only what M adds beyond G is charged.*
- *$L(D \mid M, G) = -\sum_{t=1}^n \log_2 p_M(x_t \mid x_{<t}, G)$ is the idealized predictive codelength of D under M given G (Lemma 1).*
- *The total description length, conditional on G , is $DL(M, D \mid G) = L(M \mid G) + L(D \mid M, G)$.*

Definition 2 (Turing gain). *Turing gain operationalizes one measurable component of understanding: net held-out explanatory compression. Let B be a declared baseline explanation, M a candidate explanation, D a sequence of observations that were not available when M was constructed or selected (sealed held-out data; see Section 5), G the declared, versioned, cost-accounted shared background (pretrained models, corpora, interpreters, toolchains; Section 5.7), and P a measurement*

profile satisfying Definition 3. The profile may record G within P , but G appears in the notation explicitly: every codelength is conditional on the shared background, and the conditioning set travels with the score. The Turing gain is

$$\mathcal{T}(M; B, D, G, P) = DL(B, D \mid G) - DL(M, D \mid G)$$

equivalently,

$$\mathcal{T}(M; B, D, G, P) = [L(B \mid G) + L(D \mid B, G)] - [L(M \mid G) + L(D \mid M, G)].$$

Two reporting kinds are distinguished. $\mathcal{T}_{\text{ideal}}$ is the complexity-adjusted held-out log-loss improvement: the net description-length difference computed from idealized predictive codelengths, requiring no actual entropy coder. $\mathcal{T}_{\text{real}}$ is the complexity-adjusted improvement in an actual reference-coded bitstream: the net difference between the bit lengths produced by coupling B and M to the profile's declared reference coder on D . Hypothesis H1 tests whether $\mathcal{T}_{\text{ideal}} \approx \mathcal{T}_{\text{real}}$ within the profile's declared bound for the tested regime. Until H1 passes for a regime, idealized results in that regime may not be reported as $\mathcal{T}_{\text{real}}$; they are reported in $\mathcal{T}_{\text{ideal}}$. The **reporting unit** is the Turing, T : one Turing is one bit of net held-out explanatory compression under a declared protocol. A **measurement** is a reported value together with its profile (e.g., 84,220 T under profile P). $\mathcal{T} > 0$ means the candidate's total description length is smaller than the baseline's, whether through better prediction, lower model cost, or both; $\mathcal{T} = 0$ means no net explanatory gain; $\mathcal{T} < 0$ means the candidate's total description length is larger, through excess complexity, worse predictions, or both. A smaller candidate can score negative through worse predictions alone, and positive gain can come from a cheaper representation with identical predictions. The **specific Turing gain** is $\mathcal{T}(M; B, D, G, P)/n$ for $n = |D|$ sealed held-out observations, in Turings per observation; it is the headline reported number, and raw totals are compared only at fixed n . Writing the gain per observation makes its n -dependence explicit: $\mathcal{T}/n = [L(D \mid B, G) - L(D \mid M, G)]/n + [L(B \mid G) - L(M \mid G)]/n$. The model-cost term decays as $1/n$, and the predictive term need not be stationary in n , so specific gain is a useful normalization for comparison, not a fully sample-size-independent constant. The profile additionally reports $\mathcal{T}$ under two to three reasonable alternative baselines (**baseline_sensitivity**), so that a gain cannot be an artifact of a weak B .

Definition 3 (Measurement profile). A Turing measurement is reported only together with a measurement profile: a versioned record specifying at minimum the fields in Table 1. Gains are comparable only under the same profile version. A bare number of Turings without a profile is scientifically incomplete.

The profile has two layers, and they must not be confused. The *comparison protocol* states what is scored and how: the observation source, the split, the baseline, the candidate-encoding and model-cost codes, the reference coder, the codelength method, the probability representation and quantizer, the evaluation procedure, and the sensitivity analyses. The *execution manifest* states how the scoring was physically performed: software versions, the machine and hardware on which it ran, the energy instrumentation and its calibration, and the attribution rule. Gains are comparable only under the same comparison protocol. H5-style cross-hardware comparisons and reimplementations deliberately vary the execution manifest while holding the comparison protocol fixed; this does not contradict the identical-profile requirement, because the requirement binds the comparison protocol, not the machine it runs on.

Definition 4 (Turing yield and Turing rate). Let E_{eval} be the energy in joules consumed by the evaluation run that produced M 's predictions on D , measured by the profile's energy instrumentation

and attributed under Definition 5. For discovery experiments, let E_{search} , E_{train} , $E_{\text{synthesis}}$, and E_{verify} be the measured energies of the discovery phases. Then:

$$Y_{\text{eval}} = \frac{\mathcal{T}(M; B, D, G, P)}{E_{\text{eval}}} \quad \text{and} \quad Y_{\text{discovery}} = \frac{\mathcal{T}(M; B, D, G, P)}{E_{\text{search}} + E_{\text{train}} + E_{\text{synthesis}} + E_{\text{verify}} + E_{\text{eval}}}.$$

Y_{eval} is the evaluation yield. $Y_{\text{discovery}}$ is the discovery yield, and it is the stronger quantity: “understanding gained per joule burned” must include the cost of discovery, since a model might require 10 MWh to discover and 1 kJ to evaluate, and reporting only evaluation yield would misrepresent its efficiency. Let s be the wall-clock duration of the evaluation run in seconds; $\mathcal{T}/s$ is the Turing rate. None of these asserts that energy has been converted into intelligence in any thermodynamic sense; they state that under the given experiment and profile, the computation produced the stated bits of verified explanatory gain per unit of physical resource consumed. Note the fit: AIEN/Omega is designed to account for synthesis, verification, and realization costs rather than treating evaluation as the whole computational process. The $Y_{\text{discovery}}$ denominator requires exactly that accounting. Physical energy instrumentation exists: the AIEN/Omega R15 performance harness completed its qualification on the DGX Spark (16 of 16 gates PASS, merged as pull request 67 on 2026-09-29, with package energy measured per operation in the machine-readable receipt). The first profile-bound idealized Turing result has now been recorded separately (pull request 85: a specific gain of about $0.44 T_{\text{ideal}}$ per observation, totaling $+2,559,679.825 T_{\text{ideal}}$ over about 5.8 million observations, under measurement profile V0; Section 7), but the R15 energy measurements have not yet been joined to a Turing-gain receipt; no yield figure (T/J) is reported until that joining exists.

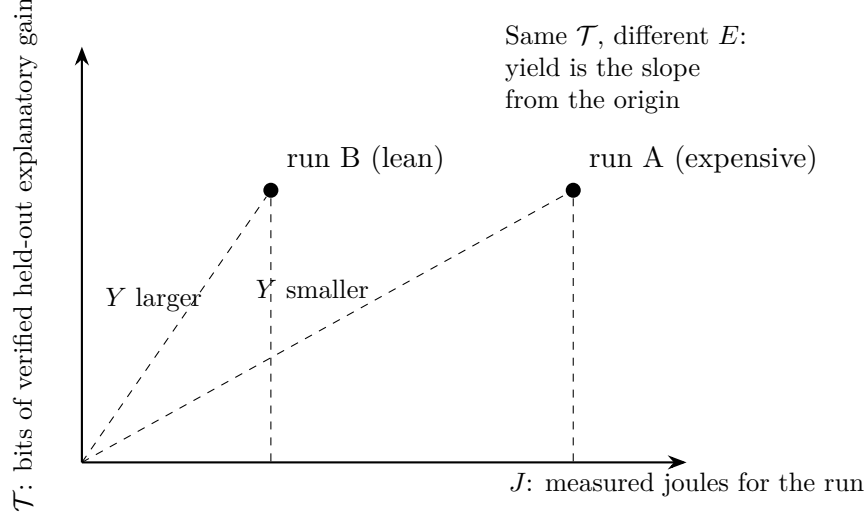


Figure 1: Conceptual illustration of the distinction between Turing gain $\mathcal{T}$ and Turing yield $Y = \mathcal{T}/E$ (Definition 4). Two runs can produce the same net held-out explanatory gain while consuming different measured energy; the yield discriminates between them, which is the environmental hypothesis H5. Conceptual illustration, not empirical data.

Definition 5 (Energy attribution). Let E_0 be the measured baseline energy of the machine at rest, E_A the measured energy with process A running in isolation, E_B with process B running in isolation, and E_{AB} with both running concurrently. The four measurements are taken over identical

measurement intervals and matched workloads: the same task content, the same ambient conditions, and the same background services, with only the presence of processes A and B varying. Define incremental consumptions $e(A) = E_A - E_0$, $e(B) = E_B - E_0$, $e(A, B) = E_{AB} - E_0$. The interaction term is

$$I_{AB} = E_{AB} - E_A - E_B + E_0.$$

If $I_{AB} > 0$, co-execution required additional incremental energy relative to isolated execution (contention); if $I_{AB} < 0$, some cost was shared or utilization improved. A negative interaction term is a signed accounting saving, not a negative quantity of physical energy. An attribution rule assigns each joule of E_{AB} to at most one of: exclusive use, shared use, interaction overhead, or unattributed; never concealing the interaction term by assigning every joule to an individual process. The profile must declare a bound on the permissible unattributed fraction of measured energy; beyond that bound, accounting closure is uninformative and no yield may be reported.

Definition 6 (Conservation constraints). *Two accounting rules are binding on every reported measurement:*

- (Physical conservation.) *Attributed energy must reconstruct measured energy within the profile’s declared uncertainty:*

$$E_{\text{measured}} \approx E_{\text{exclusive}} + E_{\text{shared}} + E_{\text{interaction}} + E_{\text{unattributed}}.$$

A resource attribution system may not allocate more energy than was physically measured.

- (Informational conservation.) *Explanatory credit must not be duplicated. If cooperating processes jointly produce one candidate worth $\mathcal{T}$ Turings, the sum of Turings credited across processes must not exceed $\mathcal{T}$. Shared explanatory gain may remain shared; it may not be multiplied.*

4 Coding Assumptions

This section separates what is proved, what is assumed, and what is defined. The claims are kept deliberately modest: every lemma below is genuinely provable, and the assumptions are stated as assumptions rather than dressed as results.

Assumption A1 (Computable encoding for $L(M | G)$). $L(M | G)$ is the bit-length of M under a specified, *computable* encoding convention declared in the measurement profile, with the declared shared background G available to the decoder (e.g., the length of a serialized artifact plus its declared decoder, or a stated two-part code). It is explicitly *not* uncomputable Kolmogorov complexity. Consequence: Turing scores are convention-relative. Two measurements are comparable only under the same profile, which fixes the convention. This is a feature, not a defect: it is what makes the quantity measurable rather than metaphysical. (Contrast Solomonoff’s universal prior [3] and Kolmogorov complexity [4], which are uncomputable and therefore cannot serve as measurement procedures.) Each profile version blesses one of three codes for $L(M | G)$: a quantized two-part code with declared precision, a bits-back/KL code with declared prior [21], or a prequential code [19]. Where the code is underdetermined, $\mathcal{T}$ is reported under two to three reasonable codes as a sensitivity analysis. The declared prior is the standard refined-MDL solution to the underdetermined-code problem (luckiness normalized maximum likelihood [7]); for deep networks, prequential encoding has outperformed variational bounds in practice [30].

Assumption A2 (Computable predictive distribution). The candidate M must expose its predictive distribution p_M as a computable function. Where an actual entropy-coded bitstream

is not produced, the idealized codelength $L(D \mid M, G) = -\sum_t \log_2 p_M(x_t \mid x_{<t}, G)$ is used, and the profile must record which method was applied (`codelength_method`). The profile declares a reference entropy coder, arithmetic coding [20] or asymmetric numeral systems [24], with bits-back coders [21, 33] for latent-variable models; “bits” in a Turing measurement means bits under the declared coder. Idealized log-loss is realizable to within small bounded amortizable overhead by such coders, and H1 tests the bridge.

Assumption A3 (Continuous observations are quantized by declaration). Definition 1 operates on discrete observations. Where raw observations are continuous-valued, the measurement profile declares a fixed measurement precision (quantizer) $Q_{\Delta x}$, and the encoded observations are the resulting discrete measurement symbols

$$x_t^{(q)} = Q_{\Delta x}(x_t).$$

The candidate assigns probability mass to each quantization bin by integrating its predictive density over the bin,

$$p_M(x_t^{(q)} \mid G) = \int_{\text{bin}(x_t^{(q)})} f_M(x) dx,$$

and all codelengths are computed on the discrete symbols $x_t^{(q)}$ under Definition 1. Rationale: $-\log_2 f(x)$ of a probability *density* is not an invariant number of bits. It changes with the choice of measurement units and coordinate resolution, whereas $-\log_2 p_M(x_t^{(q)})$ of a probability *mass* is. The declared quantizer fixes the measurement resolution in advance, so the reported bits are literal code probabilities for declared symbols, not differential entropies. Two measurements with different quantizers are not comparable; the quantizer is part of the profile precisely for this reason.

Lemma 1 (Log-loss to idealized codelength). *For a predictive distribution p_M over a finite alphabet and a sequence D of length n , there exists a code whose length for D is less than $-\sum_{t=1}^n \log_2 p_M(x_t \mid x_{<t}) + 2$ bits, subject to the declared termination and message-length convention.*

Proof sketch. Let $P_M(D) = \prod_{t=1}^n p_M(x_t \mid x_{<t})$ be the probability the model assigns to the complete sequence. Arithmetic coding assigns the whole sequence an interval of width $P_M(D)$ within $[0, 1)$. A binary point identifying that interval requires fewer than $-\log_2 P_M(D) + 2 = -\sum_{t=1}^n \log_2 p_M(x_t \mid x_{<t}) + 2$ bits, under the profile’s declared termination and message-length convention (standard Shannon source-coding argument [2]; cf. textbook treatments of arithmetic coding). $\square$

Remark. Lemma 1 justifies reporting the log-loss sum as an *idealized* codelength. Hypothesis H1 of the experimental protocol tests whether this idealization tracks actual lossless encodings in practice; the declared reference coder (Assumption A2) is what makes “bits” literal rather than metaphorical.

Lemma 2 (Memorization bound). *Let M_{mem} be a pure memorizer: its entries are explicitly charged under profile P , and its predictions outside the memorized support S are no better than those of baseline B . On held-out D disjoint from S ,*

$$\mathcal{T}(M_{\text{mem}}; B, D, G, P) \leq L(B \mid G) - L(M_{\text{mem}} \mid G).$$

Hence if B is the same default predictor with lower description cost, the memorizer necessarily receives negative Turing gain.

Proof sketch. On D disjoint from S , the memorizer emits its default predictions, so $L(D \mid M_{\text{mem}}, G)$ is no better than $L(D \mid B, G)$ when B is that same default predictor; the gain is then

bounded by the description-cost difference $L(B \mid G) - L(M_{\text{mem}} \mid G)$, which is negative when the baseline is cheaper. $\square$

Remark. The counting argument behind the description cost (that arbitrary or worst-case tables with $|S|$ entries cost on the order of $|S|$ times a per-entry constant under any fixed computable encoding) covers incompressible tables, not every table. A table may itself contain compressible structure, and such structure is then doing explanatory work of its own. Lemma 2 is deliberately the weaker proposition: it proves only that pure memorization, explicitly charged, cannot earn positive Turings against a cheaper equal default on held-out data. That is exactly what the anti-memorization claim needs, and no more. The degenerate exceptions it names (tiny supports, pathologically bad baselines) are exactly why baselines must be declared and sane, and why negative Turings are informative rather than embarrassing.

5 Experimental Protocol

5.1 Measurement-profile discipline

Every experiment instantiates a versioned measurement profile satisfying Definition 3 (Table 1). The profile is preregistered before any sealed data is examined. The profile is also the paper’s answer to a philosophical demand: standards of what counts as an intelligible explanation shift with context, so the standard must be made explicit and versioned rather than left implicit [22]. Concretely:

1. The dataset, split, seal commitment (hash), blinding mechanism, sequence ordering, reset and warm-start rules, episode boundaries, and initial context are recorded before candidate construction or selection begins.
2. The baseline B is declared and frozen in the profile before evaluation.
3. The encoding convention for $L(M \mid G)$ and the probability representation for p_M are fixed in the profile.
4. Energy instrumentation is calibrated and its uncertainty recorded; the attribution rule (Definition 5) is stated for the concurrency regime of the run; the profile declares whether the reported yield is Y_{eval} , $Y_{\text{discovery}}$, or both.
5. After the run, all artifacts (profile, baseline, candidate, sealed data, energy traces, and the exact evaluation procedure) are published so that a stranger can recompute $\mathcal{T}$ and the reported yields.

5.2 Sealing and blinding

D must not have been available when M was constructed or selected. Two distinct guarantees are required, and they are not the same:

- **Identity (commitment).** A cryptographic hash of the held-out set D , recorded in the profile before candidate construction or selection, proves that the test set did not change afterward. Any violation discovered afterward invalidates the measurement.
- **Non-access (blinding).** The hash proves the data did not change; it does not prove that the researcher or candidate never saw it. For serious experiments, the profile must therefore name

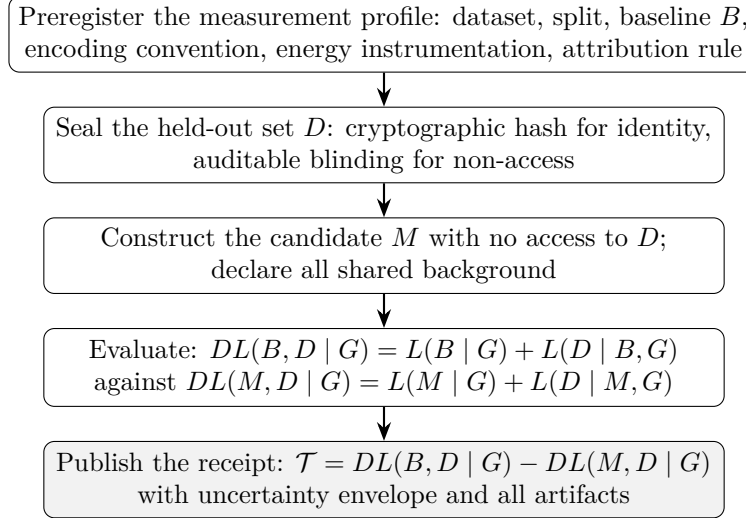


Figure 2: Conceptual schematic of the Turing measurement pipeline (Section 5). The profile is preregistered and the held-out set sealed before the candidate is constructed or selected; the reported value is the net held-out description-length difference with the candidate’s own complexity charged. Conceptual illustration, not empirical data.

one auditable blinding mechanism: an encrypted holdout released only after candidate freeze; a third-party evaluator; a physically or access-controlled test store; deterministic future-data collection after model freeze; or an equivalent. The hash proves identity; the blinding mechanism proves non-access.

Train/validation/test discipline is necessary but not sufficient. What makes the held-out claim auditable is the combination: commitment for identity, blinding for non-access.

Future data. For data that does not yet exist at the time of the freeze (fresh trajectories, future observations), hashing before construction is impossible, and the protocol does not ask for it. What is committed before the freeze is the collection protocol itself: the code and procedure that will generate the data, versioned and hashed in the profile. The realized data is hashed once collected, and the collection-protocol commitment is checked against the published code. The hash proves the identity of the realized data and the identity of the collection procedure; non-access is still established by the blinding mechanism.

5.3 Preregistered hypotheses

Each hypothesis below states a method sketch and an explicit pass/fail criterion. Concrete experiment instantiations are specified as a preregistered ladder (Section 5.4): EXP-001 is instrument validation on Omega traces, EXP-002 is Stochastic Law Discovery across three hidden stochastic worlds, and EXP-003 is active experiment selection. Each experiment is registered with a full measurement profile before data collection, and the ladder is ordered so that each rung validates the precondition of the next. Apparatus-check scripts used during protocol development are calibration of the instrument, not preregistered experiments, and their outputs are never reported as Turing measurements.

H1: Prediction–compression correspondence. *Method:* For a set of candidate models, compute idealized held-out codelengths $L(D | M, G)$ and actual lossless encodings of D produced by coupling each model’s p_M to the profile’s declared reference entropy coder. Let $\delta_M = L_{\text{real}}(D |$

$M, G) - L_{\text{ideal}}(D \mid M, G)$ be the coder overhead for M , and define δ_B likewise for the baseline; under matched model-cost conventions the two reporting kinds satisfy $\mathcal{T}_{\text{real}} - \mathcal{T}_{\text{ideal}} = \delta_B - \delta_M$. This is the test of whether $\mathcal{T}_{\text{ideal}} \approx \mathcal{T}_{\text{real}}$ within the profile’s declared bound. *Required tests*: exact lossless recovery of the encoded observations from the coded bitstream (round-trip correctness); declared per-source bounds for probability quantization, framing, termination, and other coding overhead; correct inclusion of model-description costs $L(M \mid G)$ in both $\mathcal{T}_{\text{real}}$ and $\mathcal{T}_{\text{ideal}}$; and agreement of the net gain within the bound implied by those declared overheads. *Pass*: round-trip correctness holds and absolute deviation of $\mathcal{T}_{\text{real}}$ from $\mathcal{T}_{\text{ideal}}$ stays within the profile’s declared uncertainty envelope. *Fail*: a lossy round trip, or deviation outside the declared bound. In that case the coder implementation is defective and is discarded: a failed coder implementation invalidates the implementation, not the mathematical source-coding result of Lemma 1, and idealized codelengths may not be reported as $\mathcal{T}_{\text{real}}$; they are reported in $\mathcal{T}_{\text{ideal}}$. Positive rank correlation between idealized and actual codelengths across candidates is supplementary evidence, not a substitute for bounded deviation. Until H1 passes for a regime, Turing gains in that regime are profile-bound idealized results, not measured compressions.

H2: Generalization. *Method*: Candidate abstractions produced by a discovery procedure are scored on sealed held-out D with their full $L(M \mid G)$ charged. Understanding is treated here as an ability that transfers, not a pile of propositions [29]: the profile may require scoring across multiple held-out regimes precisely to test that transfer. *Pass*: at least one candidate achieves $\mathcal{T} > 0$ with the lower bound of the uncertainty envelope above zero. *Fail*: candidates achieve positive training compression but non-positive held-out Turing gain. They have demonstrated fitting, not generalized explanatory compression.

H3: Reproducibility. *Method*: Two distinct tests. *Replay reproducibility*: the same measurement profile is re-executed with the same preregistered seed, the same artifacts, and only non-semantic re-implementations (code layout, refactors that do not change computed values). *Pass*: $\mathcal{T}$, Y_{eval} , and $Y_{\text{discovery}}$ reproduce within numerical tolerance. *Statistical robustness*: the profile is executed under different preregistered seeds drawn from the same experiment distribution (for EXP-002, different stochastic trajectories generated from the same hidden law family). *Pass*: the conclusions remain stable within the profile’s declared statistical envelope. *Fail*: same-seed re-execution diverges beyond tolerance, or different preregistered seeds change the conclusions outside the envelope: the profile is inadequate and must be revised before any measurement under it is reported. A changed seed is not an irrelevant implementation detail: it changes the sampled data or stochastic outcome, so it tests robustness, not replay.

H4: Physical conservation. *Method*: Under concurrent workloads, attribute energy per Definition 5 and check closure against measured machine energy. *Pass*: $E_{\text{measured}} \approx E_{\text{exclusive}} + E_{\text{shared}} + E_{\text{interaction}} + E_{\text{unattributed}}$ within sensor and model uncertainty on every run. *Fail*: attributed totals consistently exceed physical totals: the attribution model is invalid and no yield (Y_{eval} or $Y_{\text{discovery}}$) may be reported from it.

H5: Efficiency discrimination (the environmental hypothesis). *Method*: Take semantically equivalent realizations of the same explanatory gain (same $\mathcal{T}$ within tolerance on the same D , G , and profile) across different algorithms, representations, or substrates, and compare Y_{eval} and $Y_{\text{discovery}}$ where discovery energy is measured. *Pass*: at least one pair of equivalent realizations exhibits materially different yield outside the uncertainty envelope. *Fail*: yield never discriminates between realizations: in which case Turing yield adds no information beyond $\mathcal{T}$ alone for the tested regime. *Interpretation*: this is the hypothesis of Section 2.1 in falsifiable form. A pass makes Turing yield an instrument for steering toward informational productivity per joule; a fail bounds its usefulness and states the limit plainly.

5.4 Planned experiment instantiations

The hypotheses above are tested through a preregistered ladder of experiments. The ordering is deliberate: the first rung asks only whether the instrument works, the second whether it can detect discovered stochastic structure, and the third whether the machine can choose the experiment that most efficiently distinguishes competing explanations.

EXP-001: Instrument validation (Omega traces). Real Omega search and synthesis traces are divided, before candidate construction, into train, validation, and sealed test. Simple predictors (uniform, empirical frequency, a small Markov model, an existing fixed heuristic, and a candidate explanatory abstraction) are compared with full total description length charged. *Pass*: stable, reproducible Turing measurements with $\mathcal{T} > 0$ achievable by a candidate abstraction that earns its complexity, and the lower uncertainty bound above zero (H1–H3). This rung asks only: can $\mathcal{T}$ be measured reproducibly at all?

EXP-002: Stochastic law discovery. A hidden evaluator generates quantized trajectories from a stochastic process whose governing equation, family label, parameters, and random seed are unavailable to the candidate; Assumption A3 governs the quantization, and the quantizer is fixed in the preregistered profile. The candidate receives development observations and may construct a computable predictive explanation M . After candidate freeze, fresh held-out trajectories are generated under an auditable blinding protocol. The candidate is compared against preregistered weak, deterministic, and stochastic baselines using full total description length $L(M | G) + L(D | M, G)$. Initial hidden worlds test pure diffusion, diffusion with drift, and mean-reverting dynamics. A successful candidate must improve sealed predictive description length after paying its own complexity while maintaining calibrated uncertainty; adding unsupported structure must not receive positive explanatory credit. Where feasible, held-out trajectories include interventional regimes: the candidate must predict under interventions, not merely under passive observation, because the best finite-data predictor need not recover causal structure [27]. Explanations that compress multiple hidden worlds earn the sum of their Turing gains, rewarding unification across phenomena rather than narrow fitting.

The three hidden worlds, with explicit falsifiable expectations:

EXP-002A: Pure diffusion, $dX_t = \sqrt{2D} dW_t$. The candidate should discover that the increments are centered and that predictive uncertainty grows with elapsed time. Crucially, it must not earn Turings by inventing deterministic curves inside random fluctuations: a polynomial fit to a short random trajectory may compress development data. On any finite sample an overfit curve can score positively by chance, so the preregistered test compares polynomial fits against an otherwise matched zero-structure candidate across repeated null worlds, with a false-positive criterion (Lemma 2’s discipline applied to overfitting).

EXP-002B: Diffusion with drift, $dX_t = \mu dt + \sqrt{2D} dW_t$. The test is whether adding the drift parameter μ earns its complexity on unseen trajectories. When the hidden world has $\mu = 0$, the preregistered test compares the drift-augmented candidate against an otherwise matched zero-drift candidate, with a false-positive criterion across repeated null worlds. Chance fluctuations can favor an unnecessary parameter, and against a weak baseline a candidate can score positively by capturing diffusion regardless of its drift term. Structure without support pays the complexity charge without repayment; whether it earns no net credit is decided by the preregistered false-positive test across repeated null worlds, and no single finite sample can certify that verdict.

EXP-002C: Mean reversion, $dX_t = -\theta X_t dt + \sigma dW_t$. The system must discover a qualitatively different predictive structure (state-dependent restoring drift) rather than a quantitative tweak of diffusion.

EXP-003: Active experiment selection. With competing explanations in hand (e.g., free

diffusion versus mean reversion), the machine estimates the expected information gain of candidate interventions and proposes the most discriminating one, for instance starting farther from equilibrium or observing over a longer interval, subject to the authorization separation of Section 5.6. *Pass*: the machine-selected intervention distinguishes the competing explanations in fewer observations (or joules) than a preregistered passive baseline schedule, with the comparison itself conducted under sealed held-out evaluation.

The ladder, one line per rung:

- EXP-001 Can $\mathcal{T}$ be measured reproducibly at all?
 $\downarrow$
 EXP-002A Can a machine recognize stochastic structure without explaining noise?
 $\downarrow$
 EXP-002B Can additional structure earn its own complexity?
 $\downarrow$
 EXP-002C Can competing stochastic explanations be distinguished?
 $\downarrow$
 EXP-003 Can the machine choose the experiment that most efficiently distinguishes them?

Design note. The series is named the Stochastic Law Discovery Experiment, not “the Brownian test.” Brownian motion is the first hidden world; the framework is intended to extend to Poisson processes, Markov chains, anomalous diffusion, coupled systems, and noisy oscillators without changing the conceptual test. The series also doubles as a demonstration of this paper’s stated limit (Section 9): a false theory can temporarily compress a restricted dataset. A polynomial looks persuasive over one short random trajectory; it can still score positively on the next by chance, which is why the preregistered test uses repeated null worlds and a false-positive criterion rather than any single trajectory. The held-out complexity discipline exists to penalize overfitting systematically, not to guarantee its collapse on any one sample.

5.5 Worked illustration (not an empirical claim)

To fix the computation, suppose a baseline requires 2,600,000 bits to describe a sealed observation set. A newly discovered theory requires 18,400 bits to describe, and the same observations require 1,100,000 additional bits when encoded under the theory. Then

$$\mathcal{T} = 2,600,000 - (18,400 + 1,100,000) = \boxed{1,481,600 \text{ T}}.$$

If the evaluation consumed 30,000 J, its evaluation yield is $Y_{\text{eval}} = 1,481,600/30,000 \approx \boxed{49.39 \text{ T/J}}$. This is an arithmetic illustration of Definitions 2 and 4, not a reported measurement: no profile, no sealed data, and no instrumentation stand behind it. Since no entropy coder is involved, the illustration is stated in $\mathcal{T}_{\text{ideal}}$ terms.

5.6 From prediction to authorized intervention

Prediction alone is passive; scientific intelligence also asks *what should be measured next*. Given competing explanations H_1, H_2 , an experiment whose predicted distributions differ sharply between them carries high expected information gain, while one on which they agree carries little. A machine may therefore estimate the expected information gain of candidate interventions and propose the most discriminating one. Information value and authorization are separate quantities: a high-value experiment is a *proposal*, and whether it may be performed is decided by the authorization regime, not by its information value. This separation is a required property of all experiment instantiations under this protocol.

5.7 Shared background and the anti-gaming rule

Description length has always been relative to background knowledge; the question is never what the absolute $L(M \mid G)$ is, but what the profile declares as given, and whether it declares it completely. The most dangerous attack on the Turing is cost externalization: a tiny prompt or adapter riding on a gigantic undeclared pretrained model can score enormous net bits while the billions of bits of pretraining sit outside the measured $L(M \mid G)$ [40]. Chollet [12] made the general point: task skill can be bought with data and compute; what matters is efficiency relative to priors and experience.

The profile answers by rule, not by trust:

1. *Declare all shared side information.* Every pretrained model, corpus, tool, and prior artifact in the causal chain of producing M is versioned in the `shared_background` profile field.
2. *Score conditionally.* The reported gain is conditional on the declared background, and the conditioning set travels with the score.
3. *Forbid undeclared oracles.* Anything the explanation depends on but does not declare invalidates the measurement.
4. *Report both costs.* Where shared infrastructure is used, report the marginal task-specific cost and the amortized share of the background cost across the measurements that use it.
5. *Audit overlap.* The `overlap_audit` field names the procedure checking pretraining data against sealed held-out D ; undeclared overlap invalidates the measurement (cf. the contamination literature [39, 41, 37]).

Externalization is not always gaming. Human science does exactly this: Newton externalizes calculus. The difference is declaration. Shared background that is declared, versioned, and cost-accounted is legitimate cumulative science; undeclared background is cheating. The versioned measurement profile is the instrument that turns the loophole into a feature. In the notation of Definition 2, the declared background is G : the reported gain $\mathcal{T}(M; B, D, G, P)$ is conditional on exactly this set, and the set travels with the score.

6 Related Work

Only real, verifiable citations are included. The mathematical ancestry is credited in full; the claim of this paper is the named unit and its protocol, not the mathematics. An independent prior-art search (September 2026) across arXiv 2020–2026, the Anthropic/OpenAI/Google DeepMind research publications, and the conference literature found no existing unit named *Turing* for net held-out explanatory compression, and no equivalent named unit. The nearest neighbors are engaged below, precisely to mark the boundary.

Turing, A. M. (1950). *Computing Machinery and Intelligence*. *Mind*, 59(236), 433–460. The methodological precedent: Turing replaced “can machines think?” with an operational experiment rather than a definition. This paper follows that move, not its content; the distinction between the imitation game and the Turing unit is drawn in Section 10. Nothing here claims Turing’s endorsement. The evidence must stand independent of personal identity.

Shannon, C. E. (1948). *A Mathematical Theory of Communication*. *Bell System Technical Journal*, 27, 379–423, 623–656. Establishes the prediction $\leftrightarrow$ probability $\leftrightarrow$ description length bridge on which Definitions 1–2 rest, and the source-coding argument behind Lemma 1.

Solomonoff, R. J. (1964). *A Formal Theory of Inductive Inference*, Parts I and II. *Information and Control*, 7(1), 1–22; 7(2), 224–254. Algorithmic probability: prediction via a universal prior weighting shorter programs more heavily. The present proposal descends from this tradition but departs from it operationally: Solomonoff’s universal prior is uncomputable, whereas Assumption A1 requires a computable encoding convention precisely so that $L(M | G)$ can be measured.

Kolmogorov, A. N. (1965). *Three Approaches to the Quantitative Definition of Information*. *Problems of Information Transmission*, 1(1), 1–7. Complexity as shortest description length. Uncomputable in general; cited as ancestry, with the same operational departure as for Solomonoff.

Rissanen, J. (1978). *Modeling by Shortest Data Description*. *Automatica*, 14(5), 465–471. The Minimum Description Length principle in two-part form, $L(H) + L(D | H)$, the direct ancestor of Definition 1’s total description length. None of the MDL literature, to my knowledge, names a distinct unit for the resulting net gain; it reports the quantity in ordinary bits.

Grünwald, P. D. (2005). *A Tutorial Introduction to the Minimum Description Length Principle*. In Grünwald, Myung & Pitt (Eds.), *Advances in Minimum Description Length: Theory and Applications*. MIT Press. And **Grünwald, P. (2007).** *The Minimum Description Length Principle*. MIT Press. The modern exposition of MDL, including the connection between successful compression and the discovery of regularity that Lemma 2 formalizes for the memorization case.

Haldane, J. B. S. (1949). *Suggestions as to Quantitative Measurement of Rates of Evolution*. *Evolution*, 3, 51–56. Proposed the *darwin* as a unit of evolutionary rate (an increase or decrease of a character by a factor of e per million years). This is the naming precedent for the Turing: a scientific unit honoring a foundational thinker, denoting an experimentally constrained quantity rather than a bare physical dimension.

Hutter, M. (2006). *The Hutter Prize for Lossless Compression of Human Knowledge*, announced 2006 [9]. The prize scores the sum of decompressor size and compressed archive size on a fixed Wikipedia corpus: in this paper’s notation, $L(M) + L(D | M)$ on one public dataset, under an explicit “compression is intelligence” banner. It is the closest historical precedent for a two-part description-length score treated as a measure of intelligence. This work adds what the prize does not require: sealed held-out data, a declared and versioned shared background G , a baseline-relative net gain rather than an absolute compressed size, and the joining of measured energy to the gain (Turing yield).

Hernández-Espinosa, Ozelim, Abrahão & Zenil (2025). *SuperARC: A Test of Machine Superintelligence*, arXiv:2503.16743. The closest conceptual neighbor found in the prior-art search: it defines intelligence through “the ability of explanatory compression” with a ϕ metric. It remains distinct on every load-bearing point of this proposal: SuperARC is a benchmark score, not a named unit; it does not compute a baseline-relative net gain with the explanation’s own description cost charged; and it has no energy accounting and no sealed-data protocol. Cited as the nearest neighbor, and as evidence that the explanatory-compression framing is in the air while the unit is not.

Legg, S. and Hutter, M. (2007). *Universal Intelligence: A Definition of Machine Intelligence*. *Minds and Machines*, 17(4), 391–444. The canonical formal measure $Y(\pi) = \sum_{\mu} 2^{-K(\mu)} V_{\mu}^{\pi}$: reinforcement-learning and environment-based, with uncomputable Kolmogorov terms, no held-out data discipline, no named unit, and no energy accounting.

Chollet, F. (2019). *On the Measure of Intelligence*. arXiv:1911.01547. An algorithmic-information-theoretic account of intelligence as skill-acquisition efficiency relative to priors and experience; the denominator is experience, not joules, and no unit is proposed.

Zhang, Canlin and Liu, Xiuwen (2025). *A Theory of Machine Understanding via the Minimum Description Length Principle*. arXiv:2504.00395. Develops “to understand is to compress” as theory; proposes no unit and no held-out measurement protocol.

Anonymous (2026). *A Theory of Intelligence as Energy-Efficient Cost Reduction: Comple-*

tence, Learning, and Amortised Deliberation. Self-published manuscript, June 17, 2026, <https://amphitere.com/papers/a-theory-of-intelligence.pdf>. The closest antecedent to Turing *yield* found in the search: it defines intelligence as cost reduction per joule, but the numerator is task cost, not bits of net held-out explanatory compression; it proposes no named unit, and no author is named. Cited as an unpublished independent manuscript, to mark the boundary plainly.

Masum, H. and Christensen, S. (2003). *The Turing Ratio: A Framework for Open-Ended Task Metrics*. Journal of Evolution and Technology, 13(2). An unrelated performance-rating proposal for chess, expressed on a decibel scale relative to median human performance. Cited solely to disambiguate: it shares a name fragment and nothing else.

On “intelligence per joule” metrics. The machine-learning systems literature contains various accuracy-per-energy and performance-per-watt efficiency metrics. These measure benchmark performance per unit energy; they do not measure net held-out explanatory compression, charge the explanation’s own description cost, or require sealed data and declared baselines. The Turing is therefore not a restatement of them.

7 Empirical Results

Status: first profile-bound idealized Turing result recorded 2026-09-29 (reported in $\mathcal{T}_{\text{ideal}}$; not yet a $\mathcal{T}_{\text{real}}$ measurement).

On 2026-09-29 the first profile-bound idealized Turing result was recorded under measurement profile V0 (omega pull request 85, merged the same day). Candidate C (a table conditioning on the tried operation plus the last four outcomes) scored a specific Turing gain of about **0.44 $\mathbf{T}_{\text{ideal}}$ per held-out observation**: a total of **+2,559,679.825 $\mathbf{T}_{\text{ideal}}$** over $n = 5,834,845$ sealed held-out search events, relative to baseline B (a table conditioning only on the last outcome). The held-out events came from three seeds (8, 9, 10) never seen during model selection (seed 7 only); the pre-registered bar of 234,602 bits was cleared, with each held-out seed passing on its own. This run, TY-1+TY-2, is a completed pre-EXP instrument result with a fixed model class. It is not EXP-001: the EXP-001 preregistered runs are pending, and the H1 prediction-compression bridge (Section 5.3) remains open. The scoring tool declines to score the held-out seeds a second time: a useful procedural discipline, not proof of blinding. A separate refit check refits both models from the training files and reproduces the receipt exactly, which establishes replayability; non-access to the held-out seeds before the freeze rests on the seed discipline recorded in the profile (seeds 8–10 never seen during model selection; only seed 7 was used), not on the tool’s refusal. The complexity charge penalizes stored information that does not earn its cost on held-out observations. Under Lemma 2’s assumptions, pure memorization cannot outperform the cheaper baseline. Candidate C is itself a table-based predictor; its positive held-out gain is consistent with legitimate generalization under the complexity charge, which Lemma 2 does not forbid, subject to the limits stated below: this is an internally replayed result, and independent blinding was not established. All scoring arithmetic uses fixed-point integer micro-bits: 1 bit = 1,000,000 micro-bits (`src/turing/ty_math.c`). Probabilities are quantized to 16 bits; each symbol’s code length $-\log_2(q/65536)$ is computed with integer fixed-point arithmetic and rounded to the nearest micro-bit. Reported Turing values may therefore carry up to six decimal places: the receipt’s exact `t_microbits` = 2,559,679,824,815 micro-bits (2,559,679.824815 $\mathbf{T}_{\text{ideal}}$), displayed as +2,559,679.825 $\mathbf{T}_{\text{ideal}}$ rounded to three decimals, not a floating-point approximation. Worst-case rounding is under 6 bits over the whole held-out set. The 6-bit figure is an arithmetic rounding bound on the scoring computation, not a statistical confidence interval; statistical uncertainty for $\mathcal{T}$ is required future work. The reported baseline is a weak one (a last-outcome table); the profile’s `baseline_sensitivity` field requires the gain to be re-

ported under two to three reasonable alternative baselines, which is pending (Table 2). Attribution is likewise pending: the candidate changes operation conditioning and history length simultaneously, so the gain cannot yet be attributed to either. Required ablations are an operation-only baseline, an operation-plus-last-outcome baseline, and a longer-history-without-operation baseline; no ablation numbers are reported here (Table 2). Limits stated plainly: $\mathcal{T}$ here is a theoretical code length. No entropy coder was built that produces the smaller file, so this is not yet a $\mathcal{T}_{\text{real}}$ measurement; energy was not measured in this run, so no Turing yield (T/J) is reported. Joining R15 energy measurements to Turing-gain receipts is the next step (Definition 4).

Compliance note. This result complies with the historical profile it was scored under, measurement profile V0 preregistered 2026-09-29. It is not claimed to satisfy the current protocol in full. Put precisely: a result lacking auditable blinding is not a fully qualified measurement under the current protocol. The protocol also mandates baseline-sensitivity reporting and a stated uncertainty envelope, and the V0 run provides neither; these are recorded as pending in the ledger below. The measurement-validity criteria of Section 8, including the current statement of H1 and F1, are stated in this paper now and are applied to the V0 run retrospectively.

TY-1+TY-2 is an internally replayed descriptive result under historical profile V0. The experiment record reports separation of fitting, selection, and evaluation seeds, but independent blinding was not established. The original corpus is not publicly downloadable, so public end-to-end replay is not currently available. Statistical uncertainty, baseline sensitivity, and attribution ablations remain pending. The result therefore does not constitute full qualification under the current protocol. This does not establish contamination, and it does not make the arithmetic worthless; it distinguishes an internally reported held-out result from an independently auditable demonstration.

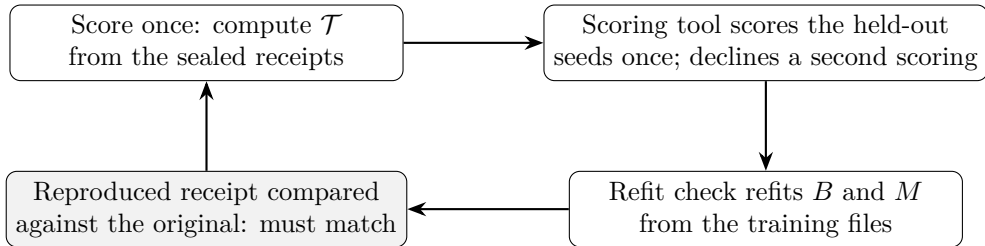


Figure 3: Conceptual schematic of the verification-by-refit loop used for the first profile-bound idealized Turing result (Section 7). The held-out data can be scored exactly once; reproducibility is established by refitting both models from the training files and reproducing the receipt bit for bit. The one-shot scoring is procedural discipline; blinding rests on the mechanism that kept the held-out seeds away from the model-selection process. Conceptual illustration, not empirical data.

Analysis plan (preregistered): for each experiment, report $\mathcal{T}$ with its uncertainty envelope, the energy attribution closure check (H4), and both yields Y_{eval} and $Y_{\text{discovery}}$ where the corresponding energies are measured; test H1–H5 per their pass/fail criteria; publish the full profile, artifacts, and energy traces for independent recomputation. Negative or null results will be reported as such. A failed $\mathcal{T} > 0$ is a result about the candidate, and a failed H3 or H4 is a result about the profile.

On prior runs. Earlier AIEN/Omega runs (including the R13 continuation run described in the companion paper *The Continuation Test*, and the R14 recovery run) predate the measurement-profile discipline and are **not scored retroactively**. Retroactive scoring under a profile that did not govern the run would violate the sealing and preregistration requirements of Section 5. The first scored run was TY-1+TY-2, conducted 2026-09-29 under profile V0: a profile-bound idealized Turing result, reported above.

8 Measurement Validity and Scientific Falsification

An immediate objection: a proposal that can fail in five ways claims nothing. The answer is that the failures belong to two different tiers, and confusing them would be the real evasion. The first tier is *measurement validity*: whether a reported number is a sound measurement under its declared profile. The second tier is *scientific falsification*: whether the quantity, once soundly measured, is a useful operational indicator of the aspect of understanding we care about. These criteria are stated in this paper now. No registration of them predates the TY-1+TY-2 result of Section 7, so that result is assessed against them retrospectively; the criteria stand preregistered for the pending EXP-001, EXP-002, and EXP-003 runs.

Measurement validity criteria. Criteria F1–F4 test the measurement implementation, the measurement profile, the T/J attribution, or the candidate. They do not test the statistic. A failure here invalidates a measurement, not the Turing.

F1: Bridge failure. If H1 fails systematically (a lossy round trip, or deviation outside the declared coding bounds), then the log-loss-to-codelength bridge is defective for the tested regime, and idealized codelengths may not be reported as $\mathcal{T}_{\text{real}}$; they remain reportable as $\mathcal{T}_{\text{ideal}}$ (Definition 2), which is exactly the status of the Section 7 result. A $\mathcal{T}_{\text{real}}$ claim survives only where the bridge holds.

F2: Generalization failure. If H2 fails (no discovered candidate achieves $\mathcal{T} > 0$ on sealed held-out data after paying its own description cost), then the proposal has not demonstrated generalized explanatory compression in the tested regime. This is a result about the candidate, not the statistic. Persistent F2 failure across diverse regimes would confine Turing gain to a theoretical quantity.

F3: Profile fragility. If H3 fails (gains vary outside the uncertainty envelope under same-seed replay or preregistered-seed variation), then the failure is in the profile: Turing measurements are not reproducible under the tested profile, and no measurement reported under that profile may be treated as a stable quantity. A unit that cannot survive reimplementaion is not a unit.

F4: Attribution failure. If H4 fails (attributed energy cannot be reconciled with measured machine energy), then the T/J attribution is invalid and no yield may be reported: Y_{eval} requires a grounded E_{eval} , and $Y_{\text{discovery}}$ requires the full measured discovery sum. Note the asymmetry: $\mathcal{T}$ can survive F4 while both yields cannot.

Scientific falsification criteria. What can ultimately fail is the scientific claim: that the net held-out description-length difference $DL(B, D \mid G) - DL(M, D \mid G)$ is a useful operational indicator of the aspect of understanding we care about, the discovery of compact structure that continues to predict unseen observations. The claim fails if sound, reproducible, profile-bound measurements of $\mathcal{T}$ repeatedly fail to track any independent sign that the machine has learned something reusable: large measured gains that never predict success on genuinely new phenomena, or a statistic that consistently rewards artifacts no independent test of understanding corroborates. Then the number would be a valid measurement of something, but not of what this paper claims. That is how the proposal dies as science while its arithmetic survives.

The test is made concrete as follows. Preregister a set of transfer tasks distinct from the scored held-out data: prediction on genuinely new phenomena (new hidden-world draws in EXP-002, interventional regimes of Section 5.6) and downstream intervention tasks with preregistered expected answers. For each candidate, record three quantities: its $\mathcal{T}$, its raw held-out log-loss improvement (without the model-complexity charge), and a naive complexity proxy such as parameter count. The proposed interpretation predicts that $\mathcal{T}$ predicts transfer-task performance *better than* raw held-out log-loss improvement alone predicts it: candidates whose gain survives the complexity charge should transfer where candidates whose improvement was merely log-loss reduction do not. The

comparators are deliberately simple: raw held-out log-loss improvement, and log-loss improvement penalized by naive parameter count. The interpretation counts against itself if, across the preregistered transfer tasks, $\mathcal{T}$ predicts transfer no better than raw log-loss improvement does, or if large measured gains systematically fail to predict transfer success while smaller log-loss improvements succeed.

F5: Supersession. If a better measure of verified explanatory gain is proposed (one that the Turing cannot match on the criteria of this paper), the Turing should be superseded. A measure of understanding should itself be exposed to falsification.

Partial failure is informative: for example, F4 failing while F1–F3 hold would establish the Turing as a valid informational measure whose physical pairing needs better instrumentation, not as a failed proposal.

9 What the Turing Does Not Measure

Following Turing’s procedure of meeting objections in turn, the limits are stated plainly. The Turing measures understanding-as-explanatory-compression: the epistemic dimension on which physics, statistics, and the minimum-description-length tradition agree. The constructive-empiricist reading [18] is the natural philosophical home of the claim: the metric scores empirical adequacy per unit of theoretical cost, a genuine scientific virtue even where the best-compressing model is not the true one [27]. The Turing must not be presented as measuring all of intelligence, nor as a certificate of safety or virtue. It does not directly measure: consciousness; subjective experience; moral worth; creativity in every sense; social understanding; agency; safety; wisdom; authority; or truth independent of the evaluation regime.

The exclusions are not an afterthought; they are what make the quantity crisp. The Turing does not measure *causal direction*: the covering-law tradition once identified explanation with predictive derivation [17], and its counterexamples are the reason this paper does not. It does not measure *semantic grounding*: a system can predict strings superbly with no model of what they are about [34]. It does not measure *agency*: passive compression captures only some of the abilities intelligence requires, while acting in the world additionally demands planning, perception, and intervention [26]. And it does not measure *value alignment*: the metric scores epistemic virtue, how efficiently regularities are captured, and is silent on which regularities are worth capturing and what the understanding is for [35]. A complete account of machine understanding needs all four; this paper supplies the one that can currently be measured with a number.

To the challenge that any formalization must say which real-world phenomenon it captures [42]: the phenomenon is theorizing itself, the cumulative practice by which expensive, weakly predictive accounts are replaced with cheaper, more predictive ones. That is narrower than intelligence, and fully within what the Turing measures.

A false theory may temporarily compress a restricted dataset. A deceptive model may predict humans well. A trivial environment may be highly compressible. A specialized predictor may score well while possessing little general competence. A well-calibrated predictor may still be misused. Consequently, verification, experimental design, distribution shift, causal intervention, and adversarial evaluation remain indispensable, and nothing in Sections 2.2–2.3 should be read as claiming otherwise: the Turing contributes a falsifiable evidentiary criterion for one operational component of understanding: not a sufficient condition, and not a safety guarantee.

10 Relationship to the Turing Test

The Turing Test asks, operationally, whether machine behavior can be distinguished from human behavior under an interrogation procedure. It is a finite game with a binary verdict, and it measures *imitation*: can the machine appear intelligent to a judge, for the duration of the interrogation?

The Turing unit asks a different question, in a different kind of game. Not “can the machine imitate us?” but “**can the machine discover structure that makes reality more predictable, and can that structure pay for its own complexity on observations it has never seen?**” It is an infinite game with no final verdict: gains are reported per experiment, under declared profiles, indefinitely, as the system continues. A high Turing gain does not imply passing any imitation test, and passing an imitation test implies nothing about Turing gain. A fluent memorized performance can pass the former while scoring at or below zero on the latter (Lemma 2, under its stated assumptions: a pure memorizer, explicitly charged, against a cheaper equal default on held-out data disjoint from its memorized support).

The contrast with benchmark culture is deliberate. A benchmark asks: how accurately does the system perform on this test set? A Turing measurement asks: how much sealed, unseen reality does the explanation make less surprising, after paying for itself, with the background declared? The first rewards performance, which can be bought with data, memorization, and prompt engineering [12, 37]; the second rewards explanatory efficiency, which cannot be bought without being charged. A large language model is a candidate under this protocol like any other: it may score Turings, provided its pretraining is declared as shared background and its held-out data is provably held out (Section 5.7). The protocol does not prejudice the architecture; it audits the accounting.

The latter does not replace Turing’s imitation game. It extends the same methodological instinct (replace the unanswerable question with a measurable one) into a domain in which modern machines can be evaluated continuously and quantitatively:

$$\text{imitation} \rightarrow \text{prediction} \rightarrow \text{explanation} \rightarrow \text{experiment}.$$

11 Conclusion

Turing transformed the question of machine thought by declining to settle the meaning of thought first. This proposal follows that methodological example. There is no need to agree on a complete philosophy of machine understanding before beginning to measure one of its components.

I begin instead with a smaller experimental question: *has the machine discovered an explanation that is simpler than what came before, survives contact with unseen reality, and reduces what remains surprising?* If it has, the reduction can be measured. I call one bit of such net, verified, held-out explanatory compression **one Turing**.

The proposal may fail, and Section 8 states exactly how failure will be recognized. That falsifiability is a requirement of the proposal, not a defect.

If it succeeds, even partially, it supplies three things the field currently lacks: a measure that makes the energy cost of machine understanding legible and comparable as compute scales; a discipline that rewards verified explanatory gain over fluent performance; and a measurable target for the long work of building machines whose competence reflects understanding rather than performance. That is infrastructure worth having: for the science, for the environment it runs in, and for the people who will live with what it produces. The claim stays narrow by design: the Turing measures understanding-as-explanatory-compression, one operational component of understanding, and Sections 5.7 and 9 state what the protocol and the quantity exclude.

The machine proposes.
The experiment measures.
Reality decides.

Acknowledgments

This paper is solo-authored. AI language models were used as drafting, analysis, and review tools in its preparation; the argument, and any errors, are the author’s own.

A Revision history

This preprint was revised through v1.14 on 2026-10-02 before release. The revision record is kept here, outside the body of the paper, for transparency.

- v1.1** Technical revision: unit/outcome distinction, Lemma 1 proof, Lemma 2 rewrite, yield split, seal/blinding distinction, claim softening.
- v1.2** Related work expanded after an independent prior-art search (SuperARC, Legg–Hutter, Chollet, Zhang–Liu, and an unpublished independent manuscript on intelligence per joule); companion paper cross-reference updated.
- v1.3** Assumption A3: continuous observations are encoded as declared quantized symbols, so Definition 1 always operates on discrete measurement symbols; new **quantizer** profile field; preregistered experiment ladder EXP-001 (Omega-trace instrument validation), EXP-002 (Stochastic Law Discovery: pure diffusion, drifted diffusion, mean reversion), EXP-003 (active experiment selection).
- v1.4** Citation corrections after source verification: Zhang/Liu author initials corrected to C. Zhang and X. Liu; anonymous 2026 manuscript given its full title, date, and stable URL; R15 state reconciled against the repository (PR 67 merged, 16/16 gates PASS, energy measured; no Turing yield yet); H3 split into replay reproducibility vs. statistical robustness; Table 2 profiles marked draft with preregistration pending; **uncertainty_envelope** profile field now requires the estimation procedure.
- v1.5** First empirical result: TY-1+TY-2 PASS (+2,559,679.825 T_{ideal} under profile V0, omega PR 85); Empirical Results section populated; Definition 4 updated to distinguish measured Turing gain from not-yet-joined Turing yield.
- v1.6** Substantive revision addressing independent review: Turing gain reframed as net held-out explanatory compression (one measurable component of understanding, with explicit scope exclusions); Turing’s ban/deciban precedent (Good 1979); specific Turing gain (per-observation rate) with totals compared only at fixed sample size; shared-background declaration and anti-gaming rule; declared reference coder; blessed model-cost code with sensitivity analysis; baseline-sensitivity reporting; benchmark-culture contrast with a fair statement of the world-models debate; physics-grounding thesis with physics-informed ML and the Bitter Lesson; interventional held-out regimes; bibliography expanded.
- v1.7** Release versioning correction: the public release restores the true revision sequence (v1.1–v1.6) after the staged website build was erroneously labeled v1.0; companion cross-reference updated to v1.7. No content change from v1.6.

- v1.8** Figures added (Sections 3, 5, and 7): the Turing measurement pipeline, the distinction between Turing gain $\mathcal{T}$ and Turing yield, and the verification-by-refit loop. Substantive revision from technical review: the $\mathcal{T}_{\text{ideal}}/\mathcal{T}_{\text{real}}$ split (the Section 7 result restated as the first profile-bound idealized Turing result); the declared shared background G made explicit in the notation, $\mathcal{T}(M; B, D, G, P) = [L(B \mid G) + L(D \mid B, G)] - [L(M \mid G) + L(D \mid M, G)]$; the measurement-profile fields extended (sequence ordering, reset/warm-start rules, episode boundaries, initial context, candidate-selection count, test-query history); falsification re-framed as two tiers, measurement validity (F1–F4) and scientific falsification.
- v1.9** Review-driven revision: TY-1+TY-2 restated as the completed pre-EXP instrument result with EXP-001 pending and H1 open; the specific gain ($\sim 0.44 \mathcal{T}_{\text{ideal}}$ per observation) made the headline; the Hutter Prize added to Related Work; the anti-memorization claim narrowed to Lemma 2’s scope; the T/tesla notation footnote added; LaTeX production fixes.
- v1.10** Review-driven revision: invalid certainties in EXP-002A/B replaced with preregistered null-world comparisons and a false-positive criterion; the H1 prediction–compression bridge restated with lossless round-trip correctness and bounded deviation as the primary requirements (rank correlation demoted to supplementary evidence), with the bridge equation $\mathcal{T}_{\text{real}} - \mathcal{T}_{\text{ideal}} = \delta_B - \delta_M$ and a stated rule that a failed coder implementation invalidates the implementation, not the source-coding result; the scientific falsification criterion made concrete with preregistered transfer tasks, expected prediction, simple comparators, and the result that would count against the interpretation; the specific-gain discussion notes the model-cost term’s $1/n$ dependence, so specific gain is not fully sample-size independent; the scoring-tool refusal wording softened to procedural discipline, with replayability from the refit check and non-access from the seed discipline; the future-data commitment rule stated (commit to the collection protocol before the freeze, hash realized data after); TY-1+TY-2 attribution ablations marked pending (operation-only, operation+last-outcome, longer-history-without-operation); the measurement profile split into comparison protocol and execution manifest; energy attribution requires matched workloads, treats a negative interaction term as a signed saving, and bounds the permissible unattributed fraction; TY-1+TY-2 evidence ledger added; repeated unit-name defenses, Turing-Test comparisons, and may-fail declarations reduced.
- v1.11** Evidence ledger completed from the live repository: exact observation count $n = 5,834,845$, per-seed gains (8: 745,936.441 T; 9: 707,551.494 T; 10: 746,835.890 T), model lengths (B: 1,156 bits; C: 180,834 bits), held-out predictive losses, Krichevsky–Trofimov smoothing, artifact and profile links with SHA-256 digests, freeze chronology, receipt identifier, and replay command; 10-candidate selection on seed 7 recorded; Table 2 headline leads with the 0.44 per-observation specific gain.
- v1.13** Final reconciliation pass. The exact micro-bit gain is corrected: the receipt’s `t_microbits` = 2,559,679,824,815 micro-bits ($2,559,679.824815 \mathcal{T}_{\text{ideal}}$), and the headline +2,559,679.825 T is that value rounded to three decimals. A TY-1+TY-2 result-status paragraph is added: an internally replayed descriptive result under historical profile V0, with independent blinding not established, the corpus not publicly downloadable, and statistical uncertainty, baseline sensitivity, and attribution ablations pending, so the result does not constitute full qualification under the current protocol; this distinguishes an internally reported held-out result from an independently auditable demonstration without establishing contamination. The Candidate C generalization claim is qualified with the same limits; the categorical “earns no net credit” verdict is softened so that the repeated null-world false-positive test carries it; the

companion cross-reference is corrected to v1.7; G (declared shared background) is defined in the Definition 1 symbol list; and “per unit of explanation cost” is restated as predictive improvement after accounting for model-description cost (the score is a difference, not a ratio).

v1.14 Figure repair: Figure 1 (Turing gain vs. Turing yield) redrawn — the y-axis label no longer collides with the axis, and the “ Y larger” / “ Y smaller” labels now sit adjacent to their respective slope lines instead of floating ambiguously.

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Field	Requirement
profile_version	Version identifier of this profile
baseline	The declared baseline B , fixed before evaluation
candidate_encoding	The computable encoding convention used for $L(M G)$ and $L(B G)$
dataset	The observation source and collection procedure
data_split	Train/validation/test partition; which portion is sealed held-out D
seal_commitment	Cryptographic commitment (e.g., hash) to D recorded before M was constructed or selected (proves identity of D , not non-access). <i>Future-data exception:</i> where D does not yet exist at freeze, commit to the collection protocol before the freeze and hash the realized data after (Section 5.2)
blinding_mechanism	Auditable mechanism establishing non-access to D before candidate freeze (Section 5.2)
probability_representation	How p_M is represented and computed
quantizer	Fixed measurement precision $Q_{\Delta x}$ for continuous observations; the encoded symbols are the resulting discrete bin indices, with bin probabilities obtained by integrating the candidate’s predictive density (Assumption A3)
codelength_method	Idealized log-loss sum or actual entropy-coded bitstream
evaluation_procedure	Exact computation steps from artifacts to $\mathcal{T}$
software_version	All software involved, pinned
energy_instrumentation	Physical measurement apparatus and calibration for energy terms
attribution_rule	How joules are attributed under concurrency (Definition 5)
uncertainty_envelope	Declared reproducibility tolerance for $\mathcal{T}$ and yields, <i>with the estimation procedure stated:</i> which of block bootstrap resampling over sealed held-out blocks (blocks rather than individual observations where observations are correlated), repeated runs under preregistered seeds, instrument/sensor calibration uncertainty, or predictive/coding uncertainty are included, and how the lower bound of $\mathcal{T}$ is computed
verification_requirements	Independent recomputation requirements
shared_background	All pretrained models, corpora, tools, and prior artifacts in the causal chain of producing M , versioned; the gain is scored conditionally on this set (Section 5.7)
reference_coder	The declared entropy coder coupling p_M to bits; “bits” means bits under this coder (Section 4)
model_code	Which $L(M G)$ code is blessed: quantized two-part, bits-back/KL, or prequential; sensitivity analysis where underdetermined
baseline_sensitivity	$\mathcal{T}$ reported under two to three reasonable alternative baselines besides the declared B
overlap_audit	Procedure auditing pretraining data against sealed held-out D ; undeclared overlap invalidates the measurement
sequence_ordering	The order in which the observations of D are scored; $p(x_t x_{<t})$ depends on it, so the ordering is fixed in the profile before evaluation
reset_warmstart	Reset and warm-start rules: what state, if any, carries between sequences or episodes
episode_boundaries	How episode boundaries are defined and whether predictive context crosses them
initial_context	The context available to the candidate at the start of each held-out sequence (prior observations, prompts, system state)
selection_count	How many candidates were constructed or compared before the reported one
test_query_history	Whether the sealed test set, or any part of it, was ever queried by any earlier candidate or analysis; any such query invalidates the seal

Table 1: Required fields of a Turing measurement profile.

Expt.	Workload	Baseline	Profile	$\mathcal{T}$ (T)	E_{eval} (J)	$E_{\text{disc.}}$ (J)	Y
TY-1+TY-2	Omega search traces (held-out seeds 8–10)	last-outcome table	profile V0	0.44/obs.; +2,559,679.825 total	<i>pending</i>	<i>pending</i>	<i>pe</i>
TY-1+TY-2			<i>alternative baselines: pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pe</i>
TY-1+TY-2			<i>attribution ablations: operation-only, operation+last-outcome, longer-history-without-operation</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pe</i>
EXP-001	Omega traces (instrument validation)	uniform / empirical / Markov / heuristic	profile v1.0 (draft)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pe</i>
EXP-002	Stochastic law discovery (diffusion, drift, mean reversion)	weak / deterministic / stochastic	profile v1.0 (draft)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pe</i>
EXP-003	Active experiment selection vs. passive baseline	passive schedule	profile v1.0 (draft)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pe</i>

Table 2: Results. The TY-1+TY-2 rows report the first profile-bound idealized Turing result (2026-09-29, profile V0), in $\mathcal{T}_{\text{ideal}}$; the headline is the specific gain, about 0.44 T_{ideal} per held-out observation. Profile fields for the run: **selection_count**: 10 candidates compared on seed 7 (fit seeds 1–6), reported candidate is Candidate C; **test_query_history**: sealed seeds 8–10 were never queried by any candidate or analysis before scoring. The alternative-baselines and attribution-ablations rows are pending. EXP rows await runs under preregistered profiles. Numeric cells marked “pending” await sealed runs with energy instrumentation.

Evidence item	Recorded value or status
Held-out observations	$n = 5,834,845$ sealed held-out search events (seeds 8–10)
Per-seed gains	Seed 8: 745,936.441 T over 1,969,906 events; seed 9: 707,551.494 T over 1,884,427 events; seed 10: 746,835.890 T over 1,980,512 events. Each per-seed figure charges the full 179,678-bit model-cost difference once, so the three sum to 2,200,323.825 T; the pooled total charges it once overall, hence $2,200,323.825 + 2 \times 179,678 = 2,559,679.825$ T. Each seed passed the 234,602-bit bar on its own.
Baseline B	Table conditioning only on the last outcome; model description length 1,156 bits
Candidate M	Table conditioning on the tried operation plus the last four outcomes; model description length 180,834 bits
Predictive losses	Exact held-out log-loss: $B = 7,482,868,867,446$ micro-bits; $M = 4,743,511,042,631$ micro-bits. The printed three-decimal values (7,482,868.867 and 4,743,511.043 bits) are each rounded independently from these exact micro-bit values. The exact gain is the receipt's $t_microbits = 2,559,679,824,815$ micro-bits, i.e. 2,559,679.824815 T; the headline +2,559,679.825 T is that value rounded to three decimals.
Probability representation	Probabilities quantized to 16 bits ($q/65536$); fixed-point integer micro-bits (1 bit = 1,000,000 micro-bits); arithmetic in <code>src/turing/ty_math.c</code>
Smoothing	Krichevsky–Trofimov (add-1/2) smoothing, quantized to 16 bits: every table row adds one half-count to each cell, rows sum to exactly 65,536, every probability $\geq 1/65536$
Serialization	Model and scoring arithmetic in <code>src/turing/ty_math.c</code> ; artifacts pinned at merge commit <code>f0dc0af9db663bd36bcafe66c3d5ac79bbf143e5</code> : https://github.com/aien-dev/omega/tree/f0dc0af9db663bd36bcafe66c3d5ac79bbf143e5/evidence/TURING_YIELD/ . The trace corpus lives outside git at <code>~/aien-data/crumblin</code> and is not publicly downloadable.
Blinding mechanism	What exists: reported seed discipline. Ten candidates were compared on validation seed 7 (fit seeds 1–6); seeds 8–10 were never seen during model selection, as recorded in the 2026-09-29 selection/freeze record, omega PR 85, and the verify log; the scoring tool declines to score the held-out seeds a second time (procedural discipline, not proof of non-access). What does not exist: independently auditable blinding. No separate access-control or access-log mechanism guards the held-out traces; non-access rests on the recorded seed discipline, not on an audited barrier.
Freeze chronology	2026-09-27: trace corpus generated (seeds 1–10). 2026-09-29: profile V0 pre-registered; scoring run executed; independent verify run 22:00:05 UTC, VERIFY OK. Merged 21:05:27 UTC as omega pull request 85 (full merge commit <code>f0dc0af9db663bd36bcafe66c3d5ac79bbf143e5</code>).
Measurement profile	Profile V0 (<code>turing.yprofile.v0</code> , SHA-256 <code>f4292df9748e306165f30b250673a0062391054321ea6dbe68855ecaa0fa8be8</code>); human-readable at https://github.com/aien-dev/omega/blob/f0dc0af9db663bd36bcafe66c3d5ac79bbf143e5/docs/turing/TURING_YIELD_PROFILE_V0.md . Not the EXP-001-track <code>calibration/profiles/Turing-profile-v1.0.toml</code> .
Receipt	<code>evidence/TURING_YIELD/ty2_heldout_receipt.txt</code> (SHA-256 <code>11df23e1a213f408a74d2c9303643b8e2f29ce2c6a51fe3f59ccc1e168401f66</code>), pinned at https://github.com/aien-dev/omega/blob/f0dc0af9db663bd36bcafe66c3d5ac79bbf143e5/evidence/TURING_YIELD/ty2_heldout_receipt.txt ; verdict PASS; gain-record digest <code>823cd3a6589218cd5e1550fe8a61ac97ca539e289fd89f3f7397ded99e9e5af6</code> ; the receipt's exact $t_microbits = 2,559,679,824,815$
Replay command	From the verify log: <code>make turing-verify && build(tests-turing-verify/turing-verify</code>